

Class Lectures in Language Acquisition

Preface & Prologue

These class lectures on Child Language Acquisition included a pedagogical treatment of the so-called ‘Four Sentences’, along with an ‘Accumulative Lecture: Form defines Function’—both of which present insights into traditional questions dealing with ‘learnability problems’ associated with language acquisition. These lectures present an overview of the defining notion that language is quintessentially an internal mental processing; that it is the internal representations of our minds (Form) which define our environments (Function). This is indeed what we find of language—that language is largely a private enterprise, which, in fact, bears very little to what one would expect of a mere ‘channel for communication’.

The unique properties which govern language are an outlier, they are a series of ‘black-swan’ events.

Syntax shows us such sweeping powers of recursive complexity that it becomes quite difficult to ponder the exact nature of its origins. Certainly, a Darwinian-evolutionary tale is not without its problems in this respect, given that what we see of the formal properties of human syntax is mostly devoid of mere communicative aims.

Other general topics in the lectures include matters related to Child Syntactic Development, Second Language (L2) issues also accompany various discussion points as a means to contrast L2 from first language (L1). Regarding L2 phonology, students will enjoy the ramifications of so-called ‘Phonological Repair’ when looking at English borrow-words in Japanese—e.g., how ‘love story’ might get pronounced as ‘loba sutori’, or ‘taxi’ as ‘takushi’, etc.

Prologue

A Brief History of Psychology. Let's begin with some interesting and historical analogies related to (i) the technology-interface to learning, and (ii) brain-analogies. It's interesting to question what the many psychological impacts have been on the state of our human evolution. For instance, we can start with the invention of *paper* and what its lightweight and easy transport has meant for the establishment of learning. (I am reminded of the discovery of the Archimedes Palimpsest, the tenth century manuscript found when its original *vellum*, the dried animal skin used before paper, revealed what was just underneath its surface—as monks, three centuries later, recycled earlier vellums by *scraping-off* the prior script. We can only be horrified by the sheer volume of writing lost over time). Of course, the typical inventions follow—all of which bring very different psychological impacts: the (*movable type*) *printing press* and how the eventual spread of knowledge (sciences, religion) played on our human psychology. The *typewriter*, the *PC computer*, *advancing software* (the ability to cut & paste and copy), the *floppy-disk*... through to all the trappings of the '*internet*' (first called the 'ethernet', and then the 'information superhighway': metaphors for 'ethereal & otherness' (the neither 'here nor there'), and of unfathomable 'speed'). These innovations are often reduced and treated as 'hardware' developments, as artifacts—but it is indeed interesting to ask, in retrospect, what such incremental progress meant for our human psychology, what it meant for our human, biologically based 'software' (i.e., the human mind). It is instructive to look for psychological impacts and to ask how human experiences regarding an interface with such innovations have helped shape our understanding of ourselves, our fellow man, as well as the world around us.

Brain Analogies. This led to so-called (historic) 'brain-analogies'. For instance, it was once understood that the brain was analogous to mechanical 'clocks' (of the philosophical '*clouds & clocks*' argument, see Karl Popper 'Objective Knowledge: an evolutionary approach' Ch. 6). In this antiquity notion, the '**brain as clock**' was said to be made-up of levers and gears which would interact in very trivial ways with the environment. The most obvious interaction with our brain-as-clock metaphor was to count and remember things. A person was understood to be the mere product of the things we came across in our environment, the things we noticed and counted: 'man as ultimate calculator'. (See Locke's notion of man as a blank slate, a *tabula rosa*). Whether or not a person was 'smart' was based on how well he *noticed* and *remembered* his token counts of environmental interactions; of course, there was no notion as to why a person might notice one thing over another (that question might have more to do with the psychology of observation, 'a cloud'). The idea of how a 'bad' experience might impact our brain/clock (for example, how a person's personality might be affected and formed) was not considered. In fact, such ideas of 'personality' really don't begin to be formulated, psychologically, until the 19th century (coming to bear on the work of Freud, etc.) (But there were earlier antecedents for sure, found in 17th century early modern English literature: e.g., Shakespeare's first psychological profile of MacBeth).

The notion of '**brain as cloud**' begins to take shape actually in the second half of the 20th century with the work of the linguists *Noam Chomsky*, who himself brings forth 17th century arguments from the *Age of Enlightenment*'s own René Descartes in forming a '*Cartesian linguistics*'. This new 'spookiness' of the brain (a nebulous-like cloud)—an enlightenment

invention—was a large part of the underpinnings of Chomsky’s linguistics theoretical pursuits: viz., how Chomsky questions the ‘direct-environmental’ theories of language acquisition of the times. A clock account to language acquisition would have it that the child’s sole role in language acquisition is to consciously count the linguistic items she comes across in the course of her language growth. This direct ‘input-to-output’ imitation-scheme was a large attribute of the *Behaviorists* Theories behind general learning procedures (as advanced by the Behaviorists schools of thought, e.g., *B.F. Skinner*). (See **imitation to analogy to computational** below).

This classic dichotomy over **Empirical** thought (e.g., *Aristotle, John Locke, John Stuart Mill, and B.F. Skinner*) versus **Rational** thought (e.g., *Plato, Descartes, Spinoza, and Noam Chomsky*) has remained constant over time, with only the refashioning of terminology making the epic debate seem contemporary. For instance, the current developmental/psychology debate over ‘Nature vs. Nurture’, more currently coined as ‘the nature of Nature’ is nothing more than a reworking of the earlier **Skinner vs. Chomsky (1959) debate** (e.g., see Galasso 2003, as cited in Owens, 2007)¹.

The Dual Mechanism Model. This dichotomy in essence leads to how a ‘hybrid model’, (i.e., a Dual Mechanism Model), which can incorporate both clocks and clouds, might be embedded in our psychological processes of language, and how these two modes of processing indeed have a real physiological presence in our brain: viz., the idea that there are *two fundamentally different areas of the brain which bring about this dichotomy of processing*. A very simple example of this dual processing could be how an English speaker differently processes the two verbs DO in the expression: ‘How do you do?’ (noting how both verbs ‘DO’ have the same spelling and the same phonology, but how they may hold different meanings—of course, also how the two may have very different psychological realities). Try to guess which DO is a clock, and which is a cloud: (which has ‘calculative’ meaning and which is ethereal ‘neither here nor there’).

Regarding the *Dual Mechanism Model*, we can simply note how the expression ‘*How do you do?*’ allows the deletion of the first ‘do’ but not the second ‘do’ (in quick, spontaneous speech). But why might this be, given that at least on the surface-level, the two verbs appear identical?

(Note: Another example of such surface level deception is found in processing distinctions of the Spanish *Los* vs *Las*, where the former is an undecomposed memory chunk [Los] and the latter a decomposed stem+affix determiner [[La]s]—with different reaction-time processes: ‘Las’ obtains the signature of rule-based stem+affix dual processing attributed to the movement of affix {s}, while ‘Los’ (being replaced by the masculine determiner ‘el’ within the inflectional paradigm, *:*lo/el niño, vs la niña*) reduces to signal as a mere lexical item without movement: viz., where the {s} in [los] rather incorporates into the stem). Spanish speakers don’t process the masculine/plural ‘Los’ as equivalent to the feminine/plural ‘Las’ (despite their surface-level identity). There was a time when ‘lo’ was a functioning determiner

¹ Owens, Robert (2007). *Language Development: An Introduction* 7th (seventh) edition.

in Spanish, but it has since been supplanted (a form of language *suppletion*) by the *el>los* paradigm: *la>las*, *el>los*, where suppletion of ‘*el*’ has lost resemblance of paradigmatic ‘*los*’).

Regarding our ‘clock analogy’, one could say that Spanish speakers no longer count ‘*los*’ and ‘*las*’ as being two separate minutes on the *same clock*. They are on entirely different clocks: one a clock, the other a cloud.

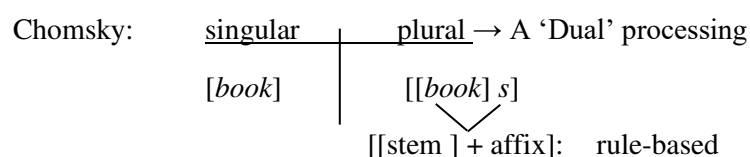
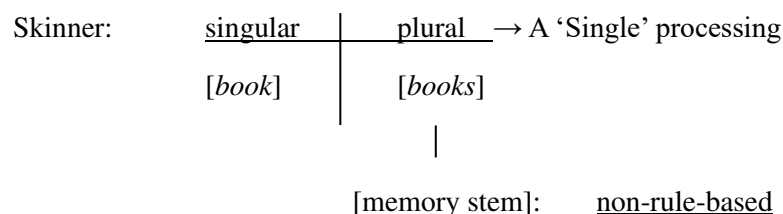
(Also see processing distinctions between very high-*frequency* regulars such as ‘walked’ over very low frequency regulars such as ‘stalked’, with high-frequency priming like undecomposed irregular-stem verbs ([*go*] > [*went*])). See Ullman, Clahsen).

A central tenet of the debate revolves around **morphological storage**: viz., how stem + affix process—whether or not stems + affixes are stored as a single item or not (the former constituting a single mechanism model, the latter a dual model). Of course, even Skinner knew that the grammatical construct for regular plural in English was *N+s=Pl* (e.g., ‘Books’). The question here is not about the *spell-out* of the grammar, per se, but rather about how such morphological grammars which hold across inflectional morphology get *processed*. (Skinner’s model on how language is learned would require that e.g., [*book*] and [*books*] are rendered as two separate stimuli, as individually coded in the environmental context by ‘singular vs plural’ respectively: the two items make-up two independent memory schemes (how *go* is memorized to change to *went*)).

Proponents for a single mechanism advocate for a **full-listing hypothesis** which has it that all lexical stems + affixes get processed (stored & retrieved) as a single lexical chunk [...]; both regular as well as irregular constructs similarly process as *undecomposed* chunks insofar that their ‘full-listing’ is directly pulled from out of the lexicon. Such memorized schemes indeed lean towards Skinnerian behaviorism since no decomposed rules are required. Such a full-listing would not only have the items ‘book’ and ‘books’ as two separate (undecomposed) lexical items, but that all derivations of a word would be likewise memorized as separate, independent items—e.g., [*speak*], [*speaks*], [*speaking*], [*spoke*]*, [*spoken*] *(noting that a dual pathway would still credit the irregular past tense ‘spoke’ as derived via an undecomposed lexical chunk).

A hybrid model, sometimes referred to as a *dual-pathway* hypothesis, more commonly known in generative circles as the **Dual Mechanism Model**, suggest that such undecomposed memorized schemes do prevail with irregulars—where rules can’t be attributed to their construct (e.g., *go>went*, *dream>dreamt*, (note the sound shift), but critically not *dream>dreamed* which would be a dual model). Many proponents of the dual model even go so far as to compromise (playing nice by debate standards) by crediting a single pathway for what appears by many to be a ‘common regular present tense verb’—for example, the verb ‘do’ seems to fall into the irregular single mechanism route when considering that the verbal present tense inflection of the verb ‘do’ undergoes the same type of sound shifts obliged by a single route of storage & retrieval: e.g., note verb-stem vowel shifts for present tense *do>does* (similar to past tense *dream>dreamt*), and, of course, the past tense *do>did* has all the hallmarks of an irregular verb. Perhaps a phonological clue here to the dual-pathway is to say that whenever there is a stem sound shift within the morphological paradigm, a new word must be realized, stored in the lexicon as a new single item.

But the hybrid dual-model provides strong evidence that decomposed rules are indeed obligatory for regular stem+affix formations (where no stem sound-shift are observed). I have suggested in various writings that the dual mechanism model can be extended to capturing the dual morphological stages of child language acquisition, where stage-1 memory of undecomposed lexical schemes come on-line prior to a stage-2 of decomposed inflectional rules. Much work of the 1960's generative grammar enterprise, including Berko's famous *Wugs test*, set out to demonstrate the nature of such creative and productive morphological processes as only decomposed rules could offer. In this sense, I often use the Skinner v Chomsky debate as a kind of 'pedagogical device' in capturing the debate on morphological processing.



This [...] edge]-feature allows for the most crucial property of language of all, that of recursion. It is this outside property at the edge which is abstract and removed from the obligatory binding of memory-based schemata. Without such an ability for *displacement, movement, affix lowering and raising, inversion, fronting*, and most crucially *embedding*, there could not be a syntax as it manifests in human language.

Let's return to the above phrase *How do you do?* Well, the two words 'DO' are not identical! In fact, the two have very different psychological states, are located in different areas of the brain, as well as hold different linguistic status. The first 'Do' is an *auxiliary verb* (a cloud) which doesn't deliver the kind of meaning usually attributed to main verbs, while the second 'Do' (a clock) is a *main verb* which delivers main verb meaning. As we can quickly see, the two verbs hold very different linguistic and psychological status: note the acceptance (in spontaneous speech) of 'How__ you do?' versus *How do you__? (where __ marks deletion of 'do'). One question that naturally follows is if the two different verbs occupy different areas of the brain, in addressing a brain-to-language corollary. They do! The discussion now leads to abstract/non-substantive functional-category words (clouds) such as the Determiner 'The', the Auxiliaries 'Do, Be, Have' (this is not an exhaustive list), etc. versus substantive/lexical-category words (clocks) such as Nouns, (Adjectives), Verbs (Adverbs), Prepositions, and how the former functional categories are attributed to Broca's area (front left hemisphere), and how the latter lexical categories are attributed to Wernicke's area (left temporal lobe). One question is how this psychological distinction plays out in the trajectory of (i) Child Language Acquisition, as well as in the production of (ii) Second language learning. Particularly, the matter of whether there's incremental

maturational onset of Broca's area for children—if clouds develop after clocks². (By extension we can similarly ask if the human Frontal Lobe (FL) of the brain (the ultimate cloud) could have only become neuro-wired later—as a ‘cascading’ consequence—after an earlier neuro-onset of the Temporal Lobe (TL). Linguistics who study ‘Proto-Language’ have this progression in mind. In any case, we know the more robust TL was neurologically connected well before the onset of a FL (both in the development of the child, as well as evolutionarily, and that earlier hominid species may only have had access to TL processes (see fn. 2)). If so, we might expect functional/Inflectional morphology to go missing in early stages of child language acquisition (2-3 years of age). This is what's behind a *maturational hypothesis* of child language.

Priming-effects & Slips. Priming effects are another example of ‘brain as clock’ since the frequency of token items seems to affect how we come to notice and remember certain language structures. We can give a couple of easy examples here:

- (i) *Word Recognition*: how the word ‘*Bread*’ (scrambled as ‘*Breda*’) seems to be unaffected when supported (top-down) by the priming of another strongly ‘associated’ word ‘*butter*’: e.g., ‘*butter*’ primes ‘*bread*’. (We also include the example ‘*Easter B__*’ to see what a word beginning with B__ might prime: *Bunny, Basket, Break, Bonnet*, are to be expected but not *Barbeque*. The notion that such priming is tied to the number of token examples as found in the input puts the brain squarely in the ‘clocks’ camp. We can also discuss how Frequency-effects (as supported by Behaviorism) led to models of Artificial Intelligence (AI) as supported by the school-of-thought called Connectionism (see link below for a paper on AI).³
- (ii) *Slips of the tongue*: For instance, why might suggest that if a possible slip is allowed, but another slip is disallowed, that this disparity might be a response to the dual mechanism model, and just questions regarding speed of processing. Consider the two slips below (one permitted, the other not):
 - a. Target: ‘The car was driving’
 - b. Slip (allowed): ‘The drive was caring’
 - c. Slip (disallowed): @‘The driving was car’.

² ‘From ‘clock to cloud’ mirrors the evolution of ‘animation’ (primate) to ‘cogito, ergo sum’ (human): a kind of ‘ontogeny recapitulates phylogeny’ argument. (See ‘Prologue note’, p. 26).

³ We'll discuss other examples, e.g., ‘Sentence No. 4’ regarding how the high frequency of the clitic form [that's] should similarly be overridden by the much lower frequency of the full form [that is], but that in fact the opposite happens—the lower frequency item wins (a very ‘spooky’ consequence which speaks to the brain as a cloud metaphor). Such examples show how frequency may not always win out in terms of a race/competitive model of processing and speaks to the brain as processing language in two fundamentally different ways (Wernicke's area rote-memorization [+Freq] vs Broca's area rule-based [-Freq]).

For my ideas on AI, see:

https://www.academia.edu/39578937/Note_4_A_Note_on_Artificial_Intelligence_and_the_critical_recursive_implementation_The_lagging_problem_of_background_knowledge_1

Priming-effects and Syntactic Constraints on Garden-paths

(See Sentence No. 3 of 'Four-Sentences' for a full discussion of garden-path sentences).

Examples of 'top-down' processing whereby language structure is employed to facilitate interpretation of sentences also shows us how language can't be simply read-off as a 'beads-on-a-string' theory and further shows how our mind doesn't work like a *clock*, but rather is dependent on abstract structure (a 'mind as *cloud*'). Consider how the sentences below become incrementally better (faster in processing) not based on the speed of 1-1 'items', but rather based on 'structure':

Examples of Priming-effects ('Top-down' processes helping with ambiguous structures):

- a. The woman painted died. (=The woman (who was) painted died)
- b. The woman painted *last week* died.
- c. The woman painted *by the artist last week* died.

Questions: How does this progression from 'less-to-more' syntax (a-c) shape our effects on processing—i.e., the more top-down syntax we have the less degree of ambiguous (d-f) and garden-path results (g-j)?

- d. The man touched Mary with silk gloves.
Who has silk gloves: The Man or Mary? This is an ambiguous sentence.
- e. The man touched Mary so *softly* with silk gloves.
- f. The man touched Mary so *pretty* with silk gloves.

Do you interpret (f) as 'Mary' is wearing silk gloves, and (e) as 'The man' wearing silk gloves? Why? Well, one might explain as follows:

The word 'Soft' primes for the verb 'touching'? (A soft touch/*A pretty touch)
The word 'Pretty' primes for the noun 'Mary'? (Mary is pretty/*Mary is soft)

- g. The horse raced past the barn __. (a fine sentence: grammatically accepted).
- h. The horse raced past the barn fell#. (#garden-path: usually not accepted)
- *(h). As soon as the final verb 'fell' is uttered, most people suffer the garden-path effect and don't want to register the verb 'fell'. However, (h) quickly gets resolved by (j) below but not (i):
- i. *The horse *that* raced past the barn. (*not a complete sentence, compare to (g)).
- j. The horse raced past the barn fell to the ground yesterday.

In (j), the preposition 'to the ground' and adverb 'yesterday' reduce the garden-path effect.

In such priming, speed of processing is untethered to the frequency of the individual item/items, *per se*, but rather is tethered to overall syntactic structure. In other words, (e-f) both words here, *soft, pretty*, as individual lexical items carry the same speed of its individual processing, but when embedded within a syntax, the items *spread-over* and act upon the overall structure in very different ways⁴. This is a ‘brain as cloud’ scenario. (See Note-3 regarding the ‘Bell-shape Curve’ in terms of this discussion on frequency, the ‘brain as clock’).

*Note in (i), by the addition of only one word ‘that’ (and a ‘non-substantive’ word to boot), we reduce in (i) the previously accepted reading of (g) and render it ungrammatical. Indeed spooky: by adding a single word to an already existing grammatical correct sentence we alter it beyond grammatical recognition. This does seem to defy mathematical logic: but then again, language doesn’t have the characteristics of a calculative math: Language is a cloud!

Slips of the tongue Errors. Such errors show a ‘language-to-brain’ corollary: viz. how the brain processes language into ‘two’ fundamentally different areas of the brain (and not as a ‘single’ clock mechanism), and how these dual linguistic properties are theoretically pegged to their relevant functional vs lexical categorial distinctions. Specifically, we note the morphological distinctions having to do with stem vs inflectional morphologies—how the verb+affix must be decomposed as [driv[ing]] and not as a holistic word/stem chunk *[driving]. If the verb ‘driving’ (and if all stem+affix formations were mere lexical incorporations) were simply memorized as a new word, then we would expect the slip in (c) to occur often: in fact, such slips/errors are completely @unattested: They don’t occur!):

Structure of slip: Note how Lexical items-[] switch with their counterpart lexical item-[], while the functional (INFL)ectional/affixes [[INFL]] remain untouched by the displacement:

- a. The [car] was [driv [ing]]
- b. The [drive] was [car [ing]] : (lexical items [car], [drive] switch and displace)
- c. @The [driving] was [car] : [driving] can’t incorporate stem+affix as lexical chunk.

Result: ‘driving’ must be processed as [driv [ing]], where only the stem [drive] is free to lexically switch with the counter lexical stem [car], but where the affix {ing} remains isolated as a decomposed affix (separate in storage & retrieval from the stem). Any morphological *full-list* theory which posits that stem+affix come fully incorporated as a ‘single lexical chunk’ [driving] doesn’t capture the evidence here: if a full-list hypothesis were correct, we would expect (c) as a potential slip. (@ But it’s unattested!).

This shows the Dual Mechanism Model behind the [stem [+affix]] decomposition—an attribute of the ‘language-to-corollary’.

⁴ ERP Experiments show that individual processing times (their ‘speed’) of lexical items and/or of processing glitches of such items show a contrast: with lexical ‘items’ being very fast N400 (400 millisecond time) while the spreading effects related to ‘syntax’ are a bit slower, e.g., P600 (600 millisecond time spans). E.g., ‘I need to butter my *socks*’ would provoke an N400, (an item-based error), not a P600.

What is the nature of this INFL-affix/outside bracket [[]]?

In theoretical linguistics (Chomsky's *Minimalist Program*, 1995), the outer bracket-[[ε]] for the structure [α, β [ε]] constitutes what is called the 'Edge'-feature* {ε}, a slot-position within the formation of both words and phrase building which houses only abstract/recursive properties of language. This recursive slot found on the 'edge' of language formation [[] ε] exclusively holds both **Functional categories** as well as **INFLectional morphology**. Their two processes both with respects to functional word-formation as well as inflectional-morphology draw unique parallels to a 'language-to-brain' corollary insofar that Edge-properties completely behave unlike their [+Frequency]-sensitive lexical counterparts (e.g., nouns and verbs). Language elements labelled as 'Edge-features' are directly attributed to Broca's area of the brain. Hence, any maturational theory of child language would have to consider the delays and onsets of such edge-properties of language, given their alignment and mapping to the maturational nature of Broca's area. (Broca's area is understood as not coming 'online' in the child until around 3 years of age).

Consider our discussion below. Note how both functional categories like the Determiner (DP) or Auxiliary verb (AUX), as well as INFL-morphology, always occupy the outer edge feature (with lexical-stem/derivational* morphology occupying the center bracket. (See below for a recap of functional/INFL elements of language).

*Note: Instances of Derivational morphology count as *new-word formations*, hence their sound-shift quality—consider the rule: 'whenever there is a sound shift, there is a new word'. And new words must get stored and retrieved as new words; they must list in the lexicon as completely separate word (e.g., (V) celebrate > (Adj) celebratory (note the vowel shift on the third syllable from /e/ to /ə/), (Adj) real > (N) reality, (sound shift on second syllable), secret > secretary, etc. These sound-shifts are a hallmark of new-word formations, as seen with Noun to Verb word change: e.g., (N) bath /æ/ (V) bathe /e/; (N) glass /æ/ (V) glaze /e/; (N) house /s/ (V) house /z/, etc.).

Keep in mind our main idea about the Clouds & Clocks metaphor and begin to question how language can't simply be a singular clock. Later we'll call this naïve assumption that language is like a clock '*a string-on-a-bead*' theory (a theory with all the trapping of Skinner's behaviorism). Below, let's recap what Lexical vs Functional categories are (along with their INFLectional vs DERivational morphology counterparts).

A Lexical vs Functional Recap:

•**Lexical Categories:** (N)oun, (V)erb, (Adj)ective, (ADV)erb, (PREP)osition.

--Irregular formations (e.g., go > went, keep > kept, write > wrote (noting sound-shifts))

--DERivational affixes: {-er} teach-er, {-ing} fascinate > fascinate-ing, {-ful} wonder > wonder-ful, {-ory} celebrat-ory

(again, noting sound shift from (V) 'celebrate' to (Adj) 'celebratory').

*Where derivational means to derive one word from another (a word-changing process: e.g. with the affix of {-er}, the verb 'teach' becomes a noun 'teach-er', etc.

•**Functional Categories:**

--(D)eterminers (e.g., *A, The, This, That, These, My*, etc.) all which merge with a potential N to form a DP (e.g., *These books*).

--(AUX)iliary Verbs (*Do, Be, Have*) along with Modals (e.g., *Can, Could; Will, Would; Shall, Should; May, Might*, etc). all which merge with a potential V to form a TP (Tense Phrase), given that AUX/edge slot provides Tense to the Verb.

--INFLectional affixes: e.g., Plural {s}, Agreement, Case (I vs me, He vs Him), Possessive {'s}, Tense [present {s}], [past {ed}], present/past participles {ing/en}.

Below, we'll consider the Determiner 'These' and the Tense feature {s} in 'Speaks'. Keep in mind that 'functional' can take the shape of either affix (e.g., plural {s}), or Word (e.g., 'These'). Derivational morphology too has its word and affix options (e.g., compounding black+bird = blackbird, break+fast = breakfast (again, noting the sound shift of *break* to BREAKfast)), with derivational affixes such as {er} (teach-er), {ful} wonder-ful), {al} critic-al, {ory} celebrat-ory.

Determiner INFL-Features (onto the N) include: Number {+/-PL}, Agreement (Number, Gender), Case (Nominative-subject (*I, He*), Accusative-object (*Me, Him*), (etc., this is not an exhaustive list).

Such INFL features:

- (i) are abstract in nature,
- (ii) define a 'language-to-brain' corollary,
- (iii) shape the recursive nature of syntax,
- (iv) provide variation among language types,
- (v) are prone to biologically-based maturation development in children (as well as being prone to critical periods in second language learning).

Let's flesh out this 'Edge' feature just a bit more below, recalling that Lexical Stems constitute in the inner [...] -bracket, while Functional/INFL constitutes the outer [[]...]-bracket:

[[Lexical stem] Functional/INFL]:

e.g., [[book] these] => [DP [D these] [N book]s]] (where there is theoretical **movement/fronting** of D to form a DP headed by the determiner 'These'—showing correct word order of D+N, since English is a *Head Initial* Language).

e.g., [[speak]s] = [TP Mary {s} [VP speak]s]] (where {s} affix **lowers** onto V stem).

What we have above is a bracket-notation which aligns ‘lexical with lexical’ and ‘functional with functional edge’ mappings (before relevant movement takes place). For instance, consider such *pre-movement* theoretical alignments:

[[books] these]: where the D ‘These’ serves two functions:

- (i) to provide a phrase-rule constituency whereby D introduces a N,
- (ii) to provides INFL plural {s} onto the Noun.

Of course, as stated in (i), English phrase-structure rules specify that D must position before the N in forming a DP (Determiner Phrase), but this *word order* is a secondary consequence of a **parameter** known as **+Head Initial** (see Lecture-1, ex. 4). There are in fact language types which show N+D: e.g., Italian *Mama Mia* (*Mother my*) where ‘My’ is a determiner in final position (a minus Head Initial position: [-Head Initial]). So, what we intend to capture here in this first instance is not so much the proper word order of the phrase, but rather how the elements within the phrase map onto what would be the inner bracket (lexical/stem) and what would be found at the edge.

A Brain-to-Language Corollary.

Albeit somewhat unorthodox and counterintuitive from what is typically shown in introductory syntax textbooks, what this mapping of such alignments shows us is how the phrasal syntactic structures map onto separate brain regions, with Lexical-[] being assigned to Wernicke’s area of the Brain/Left Temporal Lobe and the [[Edge]] (functional/INFL) being assigned to Broca’s area/Left Frontal Lobe:⁵

[Wernicke [Broca]], and where it is [... [Broca]]-area which involves movement—e.g.,

- i. [D [N, [Ø]]] in forming a DP: ‘These books’ : [[book]s]
- ii. [T [V, [F]]] in forming a TP: ‘She speaks’ : [[speak]s]

The notion of the *edge* being assigned an outer slot is tantamount to saying that its properties are on the outer margins of language, that the features contained within are formed at the periphery of language. And finally, we note that such Edge properties in fact instigate MOVEMENT, whether it be affix lowering (Verbs), Agreement between N & D (Spanish) (see below), or even ‘D-fronting’ to get to the +Head Initial position (N), as found in English.

Notice how INFL would map accordingly on N & V:

[DP These {+Pl} s [book] s] > [[N] INFL {s}] (with plural {s} moving to affix onto N).

⁵ Though of course, most images of the human brain show a left-side facing brain with Broca’s area at the upper left and Wernicke’s area at the mid-right. The order is irrelevant, only that there is a mapping of Wernicke’s area as corresponding to the inner []-bracket, and a mapping of Broca’s area as corresponding to the edge.

For instance, we know that D provides the Number [+/-Pl] feature onto the Noun by looking at D-feature agreements: e.g., while the D ‘A’ is exclusively singular in nature (*A book/*books*), the D ‘The’ allows the agreement for both [+/-Pl]: *The book/books*, etc.). An example of a determiner which specially marks for Plural would be the Determiner ‘These’ (*These books/*book*). Such properties are to be assigned as edge-features, given their abstract/syntactic qualities. Also note the Gender agreement as in Spanish DP systems: (e.g., *La Niña/*Niño*)—where both the D as well as the N have to be feminine (ending in an {-a} affix). Both Gender as well as Number are among the ‘INFL-Edge’ features which must get distributed on to their relevant lexical stems. In this sense, we know that D maps onto N and T/Aux maps on to V with regards to relevant functional/INFL features.

Noun-Mappings: ‘These books’

[Lexical [Edge]]

[N [D]]

[N book [D These]]

[N Book [D These: plural {s}]]*

[DP [D These {s} [N book]]] (= Head initial)

(before nominal affix {s} plural lowering/move)

Verb-Mapping: ‘She speaks’

[Lexical [Edge]]

[V [Aux/T]]

[V speak [Aux/T]]

[V speak [T 3P, Sg, Pres {s}]]*

[TP [T {s} [V speak]]] (=Head initial)

(before verbal affix {s} lowering/move)

[DP These [book] s] ‘These books’

[TP She [speak]s] She ‘speaks’

*(The Rule: V+{s} = 3P (3rd person), Sg (singular), Pres (Present Tense)).

*(The Rule: N+{s} =Plural (only showing the Number feature/projection here)).

Or consider how the functional Tense (T) infinitive-verb affix ‘to’ aligns with the T auxiliary-verb ‘do’ and modal ‘can’—noting that both aux verbs as well as modal, like tense, constitute an edge-feature:

(i) John likes [T to [V study]]

(ii) John [T does [V study]]

(iii) John [T can [V study]]

We notate such constructs as originating as follows (where strikethrough of edge shows erasing of edge element after movement/raising):

(i’) [to [V study [edge ~~{to}~~]]]

(ii’) [does [V study [edge ~~{does}~~]]]

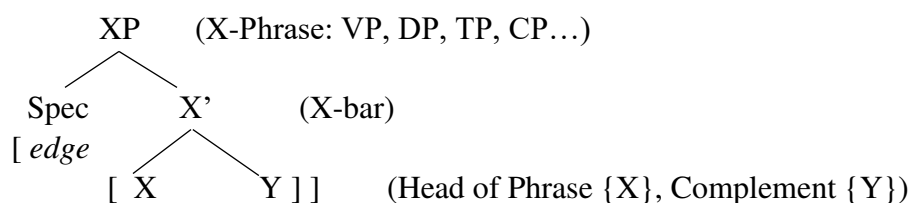
(iii’) [can [V study [edge ~~{can}~~]]]

As an extension, we consider all *functional categories* (such as the Determiner or Tense) to be an Edge-feature—despite the fact that both D and T can form the Head of their respective phrases (DP, TP). In this sense, it seems both Spec as well as Heads can be derived via an edge-feature.

This further gets extended in the theoretical-syntax literature as the distinction between +/- semantic interpretability: [+/-Interpretable] features being lexical/substantive, while the [-Interp] feature is functional/abstract in nature.

Note: (Albeit the caveat above) What's interesting about this Edge-feature is that this theoretical slot, when configured with X-bar theory, becomes the *Specifier* position which acts as an adjunct to a given Head of a Phrase. Curious enough, the Spec position is often referred to as the *Elsewhere category*. This 'elsewhere' jibes nicely with our 'brain as cloud' metaphor for 'ethereal & otherness'—viz., Spec is the '*neither here nor there*' position. This seems correct given that the Specifier position seems to be an *ad hoc* invention (attached to XP), a position seemingly prompted to serve no substantive material as would be provided by the Head of a Phrase. Rather, the position seems to have been theoretically invented for the sole purpose to host 'moved elements' from below the syntactic tree. Assuming all subjects originate within Complement of VP (*The Verb-phrase Internal Subject Hypothesis*), the subject VP *in situ* must then *raise up* to adjoin to Spec-of-TP in forming a declarative/tensed sentence. The only position free and abstract enough to host such *moved elements* is the Spec. (See Lecture-3, [§§7.1-7.3]):

X-bar



Having recapped the properties of the Edge-feature*, let's take a further look to see how such 'slips-of-the-tongue' as provided by Gordon's study shows that errors are not random but rather are quite systematic as based on our edge-mappings: where lexical errors slip and slot with other lexical areas, and where their counter functional-edge errors doing likewise.

In sum, what we attempt to show here by way of 'inside vs outside' processing-brackets is that the inner stem-formations are derived by *lexical learning*—a frequency-sensitive condition based on Stimulus & Response (S&R) methods attributed to the school of Behaviorism (e.g., B.F. Skinner). The outer edge is shown to have very different abstract qualities which form the basis of syntax. Perhaps the best and simplest way to see this is by examining Gordon's famous 'Rat-eater' experiment where children know not to insert an INFL affix (an edge property) embedded within a Noun+Noun lexical compounding e.g., rat-eater, namely, children never say for a plural form @rat-s-eater. See Gordon below for how children have tacit knowledge never to break the inherent mappings as provided by the scheme: [Lexical [Functional]], ...and that functional edge-features can never squeeze inside two lexical brackets: @[lexical [functional] lexical] as in 'rat-s-eater' where compounding of N+N {rat + eater} can't take an inserted INFL plural affix {s}, an edge-feature, between the two Nouns.

*(Edge-feature: recall, our ethereal 'neither here nor there' metaphor: Also, this same abstract slot gets labelled as the 'Elsewhere category' regarding the Specifier position within X-bar theory).

‘Rat-Eater’ Experiment (Gordon). In addition to what will be laid out in our ‘4-Sentences’ section, consider the claims made by the linguist Peter Gordon (1985) that ‘young children know not to keep plurals embedded within compounds’. In Gordon’s classic ‘Rat-eater’ experiment, children are asked: ‘What do you call a person who eats rats?’ Children respond ‘rat-eater’ (they delete the {s}) and they never respond *rats-eater⁶. Gordon suggests that children innately know that inflectional morphology {s} can’t be kept embedded within a compound, even though they have never been explicitly shown that such data is in violation of some English grammar. The mere fact that they never hear it (because it is, in fact, ungrammatical) doesn’t explain why children never entertain the prospect: children say loads of erroneous things that they have never heard before. Hence, even though children have no empirical evidence (negative stimulus) that such constructs are wrong, they still shy away from compound-embedded plurals. This is what is referred to as the **‘poverty of stimulus’**: viz. children’s inferences go beyond the data they receive.

Gordon suggests in this sense that there must be some innate built-in machinery constraining child learning of language. So, (if say strong behaviorists) ‘input-to-outputs models are the square product of environmental learning’, one question that will come up is: ‘How does such learning deliver a result such as found with the poverty of stimulus case?’ Perhaps symbol manipulation of rules will be required in some fashion after-all. But, if so, perhaps we need to rethink the brain as a mere neuro/digital network, as a super-advanced clock. In fact, the analogy of the brain as a digital computer has been under attack for some time. Perhaps at the *lower level* of []-stem-formation learning, the brain-to-computer analogy holds (where local firing of neurons takes place, etc.). But with *higher functions (Edge-features)*—when we talk of a ‘mind over brain’ in how a brain bootstraps a mind—there may need to be a fundamentally different level of processing with an entirely different neuro-underwriting. The Dual Mechanism Model does precisely this: while it treats frequency-sensitive lexical learning based on brute memorization to be of a ‘lower-level’ []-processing (attributed to the cognitive, problem-solving procedures along with Wernicke’s rote learning), it simultaneously introduces us to a ‘higher-level’ [[edge]]-processing undeterred from low-level neurological networks. This higher-level seems to be *epiphenomenal* in the sense that there is no Darwinian biological bottom-up pressure for its existence (see the Accumulative Lecture for discussion).

*Note: For Child Stage-1 **Merge**-[] operations over Stage-2 **Move**-[[]], see Endnote, Lecture-5.

We’ll come to consider the former merge operation of {x, y} as a quintessential temporal lobe/Wernicke processing over the latter move-based {x, {x, y}} frontal lobe/Broca area process.

⁶ Another great example is how the word ‘painstaking’ has suffered segmentation-metathesis from the original formation of [[pain] s]-[taking]—where the inflectional plural {s} of ‘pain-s’ finds itself as an unorthodox position in the middle of an N+N compound—to the final derivation of [pain]-[staking]. In this example, the unwanted embedded inflectional {s} gets distorted as being the initial consonant of the following word ‘taking’ (>staking).

‘Taco Experiment’ One other great study is to examine what I call the ‘Taco Experiment’...

Target: ‘What about tacos tonight?’ (the intended ‘target’ utterance of the speaker)

Slip: What about taco_ tonights? (what actually gets uttered: the slip/error)

a.Unattested: @ What about _acos tonight t?

b.Unattested: @ What about tacot sonight?

The slip we see is the result of the functional/INFLectional [plural] {s} moving to the functional edge of the word [[tomorrow]_] where there is a possible, available functional slot open—e.g., [[tomorrow]’s] (where a possessive or even a clitic {is} can be properly inserted at the functional edge. So the error is the result of the ‘same kind’, viz., a switch between functional and functional (both Broca based). However, consider why the unattested errors in a, b, can’t be made. In (a), the {t} is lexical (part of the noun stem {taco}) and can’t switch to the functional edge (* tonight t). Likewise, the ex. (b), the plural/INFL {s} can’t attached to the beginning of the lexical stem ‘tonight’ : where the unattested word ‘sonight’ erroneously mixes with initial consonant/functional element {s} attached as an initial part of the remaining lexical stem [_onight].

Clearly, if a full-listing hypothesis was in operation, something a single mechanism model (of Skinner) would want to claim, then we would expect any number of possible mistakes with no rhyme or reason—given that all words are essentially memory-chunks. Hence, if plural [tacos] was a chunk, or [speaks], or [driving], etc., any potential slips could be made in error: e.g., remove first sound/letter and move to the back of sentence, mix-up adjacent letters between two words, etc. This of course doesn’t happen. There is rather an underlying rule governing mistakes: viz., lexical errors mix with lexical positions, functional with function, as if each lexical vs functional were on their own separate and respective ‘conveyor-built’ of processing.

Vertical vs Horizontal processing. Finally, these analogies having to do with ‘clocks vs. clouds’ become overlapped with the idea of ‘vertical vs horizontal’ processing: See Note-3. Also see ms. ‘Reflections on syntax’ (link below). (For full discussion of ‘fascinating v celebrating’ types complete with syntactic tree diagrams, see page 54, link below in (iii)). Also, you can see ‘vertical v horizontal’ processing. p. 6 (link below).

https://www.academia.edu/42204713/Notes_1_3_Reflections_on_Syntax_Note_1_A_Note_on_the_Dual_Mechanism_Model_Language_acquisition_vs_learning_and_the_Bell_shape_curve_Note_2_A_Note_on_Chomskys_2013_Lingua_paper_Problems_of_Projection_Note_3_A_Note_on_Proto_Language

Class lecture topics: see link below:

https://www.academia.edu/43319709/Class_lectures_and_reflections_on_syntax_draft

(i) Introduction pp 1-41 (also begin to read lecture 3: ‘four sentences’. P 65, also link below).

(ii) Brain/mind analogies p. 12 (also brain-mind relation, p 35)

(iii) Skinner v Chomsky p 12

(iv) Vertical v horizontal (fascinating v celebrating types) p. 54. For ‘4-Sentences’ also see link:

https://www.academia.edu/37650895/Four_Sentences_Opening_remarks_for_Ling_4_17_Child_Language_Acquisition_Fall_2018

Overview of Language Structure: ‘Fascinating vs Celebrating’ Flat-[] vs. Recursive-[[]] types. Items v Categories. The Structure of Language: The ‘Four-sentences’. Evolution of Linguistic theory of Language as ‘Imitation>Analogy>Computational’.

Items vs Categories.

Consider how **items** are like ‘table, chair, nightstands’ (and represent flat-[] vertical processing), and **categories** are like ‘furniture’ (and represents recursive-[[]] horizontal processing).

[tables, chairs, nightstands [furniture]]

(See how the category (furniture) occupies the edge: [[edge]]. All abstract properties of language/syntax occupy the edge since abstraction is [-Freq] based and free to be productive). One could easily conceptualize as **substantive** *tables, chairs and nightstands*, and readily draw them, but just try to do the same for *Furniture*, draw ‘furniture’! What do you draw? (No, don’t draw a table or chair. Draw furniture! You quickly discover that you can’t since *furniture* is a categorical and abstract in nature; it is a placeholder of sorts into which tables and chairs can be placed. Using this as a metaphor, tables and chairs represent lexical items, while furniture represents an abstract quantity on a par with Edge-features of language.

See ‘Fascinating vs Celebrating’ morphology types as discussed below.

INFlectional vs Derivational Morphology and ‘Fascinating’ vs ‘Celebrating’ types:

(For full discussion of ‘fascinating v celebrating’ types complete with syntactic tree diagrams, see page 54, ‘4-sentences’ link above).

Movement distinctions based on Inflectional vs. Derivational Morphology—‘Fascinating’-types (item-based) vs. ‘Celebrating’-types (category-based).

Let’s consider the dual treatment by examining so-called [fascinating]-type processes over so-called [[celebrat]ing]-types.

But first we must give these two some structure (since any assessment of language must be structure-dependent):

(a) This is a [fascinating] class. : {ing} is ‘derivational’: V=> Adjective

(b) Mary is [[celebrat]ing] her birthday. : {ing} is Inflectional: Be + V + ing= progressive

The structure in (a) is [__ing]: [fascinating], a lexical word chunk (Wernicke’s area)

The structure in (b) is [[ing]] (*ing* is at the edge): [celebrat[ing]] (Broca’s area).

While we realize that ‘fascinating’ in (a) is an adjective, which modifies the noun ‘class’, we also see that ‘celebrating’ retains its verb status. In other words, an item-based (word) change has occurred in (a)/derivation but not so in (b)/inflection. In (a), the verb ‘fascinate’ (to fascinate) has now become an adjective (AdjP) via derivational morphology.

We can bracket the structure as ‘This is a fascinating class’ AdjP = flat [FASCINATING] (‘Vertical’ as an encyclopedic reading of items top-down) Adj N [CLASS] [fascinating] [class]. But notice how the $_{}_{ing}$ (derivational-affix) is sensitive to the [+frequency] of the item (stem): namely, the $\{ing\}$ -affix ‘Verb-to-Adjective’ derivational formation only works with a select few stems (a few select ‘items’), such as: (i) fascinate + $\{ing\}$ => Adj. (ii) amaze, (iii) interest, (iv) bore, (v) intrigue, etc.

But not (vi) * Wonder $_{}_{ing}$ (but Wonder + $\{ful\}$ (‘wonderful’).

(Note the selective properties of some verb stems which enter into adjectival derivation, all of which select their unique endings: $\{al\}$ for critic => critical, $\{ory\}$ for deposit => depository, $\{ive\}$ for collect => collective, $\{ful\}$ for wonder => wonderful, etc. etc.) Note how, for example, the verb in (vi) ‘wonder’ could not be slotted in this same $\{ing\}$ ‘verb-to-adjectival’ class: (e.g., ‘This is a *wondering class’ is ungrammatical), but that the affix $\{ful\}$ must be applied, correcting the phrase to ‘a wonderful class’. Or notice how ‘collect’ becomes collective as in ‘collective bargaining’, etc. There is a whole itemized (lexical) list of matches which stems have to consider when their relevant derivational affixes have to be paired).

This is part of the native knowledge we have in knowing our language’s lexicon—it is the kind of declarative knowledge that rides on the back of brute-force and itemized memorization schemes. Other such affix matches would be $\{ous\}$ (a dangerous road), $\{al\}$ (a critical period), $\{en\}$ (a broken window), etc. (And this is by no means an exhaustive list...It seems we just know them). But if this item-based process is completely memory-based, then, from time to time, speakers should have ‘look-up & retrieval’ glitches, a kind of bottleneck in memory, which may pose problems when recalling the correct affix match. And this is precisely what happens. Take for example the verb ‘celebrate’. How do you turn this verb into an adjective? Is it a ‘celebrating gathering?’ No! The correct affix to look-up is $\{atory\}$. (Note the phonological schwa reduction on the third stress of the stem $\sqrt{}celebrate$ from /e/ to /ə/ in celebratory).

Sometimes native English speakers stumble on this affix since it may take-on a [-Freq] ranking: ($\{atory\}$ is not often heard in the input)—e.g., we should say ‘a celebratory gathering’. So, it seems that moving from verb to adjective (an item-based derivation) is indeed rather rote-learned and frequency sensitive (i.e., reliant on the frequency amount of token examples as found in the input). In sum, what we can say about derivational morphology is that the ‘selection processes’ of affix per word is certainly ‘item-based’, as its internal processing is extremely sensitive to the +frequency of input—in the sense that specific stems must attach to specific affixes, as promoted by the frequency of such composition. In some sense, there really isn’t any decomposition at all of the [stem and [affix]], but rather the [stem + affix] seems to be processed as a completely whole new word-item with its internal semantics in place: e.g., the derivational $\{er\}$ (as in teach =>

teacher) reads off its lexical semantics as : e.g., [toaster], [driver], [dancer], [teacher], etc). So then, what about ‘celebrating types’?

Let’s consider such ‘celebrating-types’ below. Mary is [[celebrat]ing] her birthday. AspP (= Aspect Phrase: progressive ‘Be + [V +[ing]]’. The counter example here shows categorical processing (a rather horizontal mode of the processing of language). In such category-based processing, note how the affix {_ing} is rather indifferent to what the stem is (item), as long as it is a verb (category). In a sense, (quite an important sense), the inflectional {ing} is +Frequency sensitive NOT to the item/stem, but is frequency sensitive only to the category (Verb). The actual item (word) plays no role in determining whether or not the rule can be activated: all the {ing} looks for in ‘celebrating-type’ processes is to attach to a Verb-category. That is all. This is very different than what we just saw above regarding derivational morphemes, which looks-up and selects specific stem/items to attach to. We can show this rather creative productivity by placing any nonce word (a made-up word) in the V-slot and then carrying on with the rule. Try it! Say the sentence ‘John is spigulswog-ing away all of his money’ (with √spigulswog as nonce word). (Note in fact how it’s the INFL-affix {ing} itself which defines the nonce word as a verb). What we notice is that there is no look-up of whether or not the affix {ing} matches the stem-nonce. It just applied itself in all environments. This free-range of productivity is a hallmark-feature of a [-Freq of item] but [+Freq of category] processing, uniquely characteristic of MOVE.

What we find of this hallmark of MOVE is that it is rule-based (hence its productive nature): in this case, the rule would be Aspect/Progressive: Be + Verb + ing = progressive. Note how the underlying processing of such a stem and affix would inherently be decomposed as shown in structural- brackets [stem [affix]]. The form of decomposed Stem + affix, or ‘celebrating-type’ is what we term INFlectional morphology. Inflectional morphology is in stark contrast to derivational morphologies which cater to undecomposed [stem-affix] processing, a very lexical & word-based mode of retrieval, storage and recognition. While we notice how the derivational ‘fascinating-type’ {ing} is beholden (tethered) to specific stem/items, we equally note how the INFlectional ‘celebrating-type’ is free.

We can summarize the two {ing} formations below (where [-productive] requires the selection of a specific stem/item, while [+productive] is free to select any stem/item, even a nonce item):

- (a) ‘Fascinating types’ (derivational): as ‘chunking’ []. [Stem + ing] Stem is +Frequency sensitive/-Productive
- (b) ‘Celebrating types’ (inflectional): as embedded/recursive [[]].

Note in (b) how the [Stem + [ing]] is –Frequency sensitive/+Productive. However, also note that we could take the stem-verb √fascinate (to fascinate) and too turn it into a ‘celebrating-type’— e.g., ‘She is fascinating him with her eyes’ (showing Aspect/Progressive). Again, recall that all the {ing} cares about here is that it attaches to a stem (presumably a verb) in order to satisfy the progressive rule: [be + V + ing]. But a very funny thing happens when we try to go the other way, by turning the stem-verb √celebrate into a ‘fascinating type’ (i.e., by turning it into a stem-adjective): recall, how ‘celebrate’ must take on its unique [+Freq] affix {ory) as found in the structure ‘This was a *celebrating/celebratory party’ (just as we found with the example deposit >

depository cited above). Also note how the stem-internal phonology shifts—a feature very typical of a new-word classification: (celebrate) /sɛlɪbrɛt/ > (celebratory) /sɛlɪbrətɔri/. So, it seems the two affixes (Derivational vs Inflectional) psychologically behave in two different ways.

Based on these observations, let's recap how items (words) interacts with the two types of affixes and see if there are psychological conditions which place [+/-frequency] of item as the determining factor of whether or not the affix works.

Affixes are either:

- (i) Free to scan—to search for a relationship with an 'itemized' (I) stem—a search process which **isn't** beholden to a specific item (but is rather CATegory based), where [+Cat] is [-Itemized] and isn't frequency sensitive [-Frequency], isn't sensitive to specific association, but is **free**. This would have the features [-Item]/ [-Freq]. For example, the INFL plural {s} can attach to any CATegory of Noun [+Cat] and is not beholden to specific items of nouns [-I] (except in instances of irregular Nouns (man>men, mouse>mice, etc. (Note how in this sense irregulars behave like derivational [+I], [+Freq] processes. So, for INFlections, if we posit a new nonce (made-up) noun such as 'spigolit' and ask a child now there are two: they will say 'Two [[spigolit]s]. (Just like Berko's wugs text: [N [{s}]]).

Same with inflectional verbal {s}, past tense {ed}, etc. This freedom to attach to categories (such as N, V) and not to be beholden to specific items (i.e, specific word selections) is a psychological hallmark behind the ability to abstract and leads to the creative of language (Chomsky).

(Recall, Noun and Verb are 'categories'— they hold no substance quality other than how they serve as a placeholder. They are [-Freq] based since there is no memory trace of its specific item. On the other hand, the item/word like N-'book' or V-'dance' are specific lexical items [+freq] based. This dual mode is represented in our vertical [+Freq] mode vs horizontal [-Freq] mode (above)).

- (ii) Or Not free to scan. The counter to this is what happens to derivational affixes, where [+I/+Fre] determiners affixation. For instance, while the {ing} derivational affix can attach to the item-word/verb {fascinate} as in {fascinating}, deriving an adjective, (e.g., This is a fascinating class), it can't affix onto the verb wonder (e.g., * this is a wondering class), where the derivational affix {ful} gets selected (e.g., This is a wonderful/*wondering class).

Sound shifts for derivational/word change.

Clue: whenever there is a sound shift, just think: 'there is a *new* word!'

We find a complete list of such phonological stem shifts in derivational formations—for instance, just sound-out the following derivational Noun-to-Verb counterparts and notice the internal stem

sound shifts: *glass* /æ/ (N) => *glaze* /e/ (V), *grass* /æ/ (N) => *graze* /e/ (V), *bath* /æ/ (N) => *bathe* /e/ (V), *life* /ay/ (N) => *live* /ɪ/ (V), other examples include *real* > *reality*, *secret* > *secretary*, etc.

Note: This is an interesting psychological shift that takes place in the mind of a speaker in order to demarcate a different word-status.

What we are generally carrying on about here is what is referred to linguistically as MOVEMENT, and as we have seen above, there are a number of reasons for MOVE. Syntactically, the notion of movement has been expressed in GE terms by the need to either acquire and/or check-off formal features. For example, take Case (Accusative Case marker {-m} as found in the wh-object whom). Case seems to be the kind of formal feature (along with AGREEMENT) which doesn't impact the semantics of mere communication. This can be seen today by the very little observance of the accusative marker {m}—e.g., most native speakers of English prefer to say 'Who does she like?' and not 'Whom does she like?' suggesting that the case marking {-m} is rather invisible to the semantics of communication. Hence, in GE terms, such formal features (such as Agreement or Case, among other purely syntactic features) must be stripped-off of its value from the lexical item before they can enter into linguistic processing. Case as trigger for Movement Case represents one of those Syn-MOVE operations which seem to motivate movement.

'Four Sentences': (For full discussion of 'Four Sentences', see Lecture-3).

When it comes time for us to consider the classic 4-sentences (in Lecture-3), we'll closely examine how they expose the dual mechanism model of [[]]. For instance, one interesting way to tease this discussion out is to ask how a 'beads-on-a-string' theory—which is a strong behaviorist theory which posits that all words and affixes are undecomposed chunks-[], as in flat encyclopedia list of items—can't hold...just can't be right given the nature of our linguistic evidence argued (in the form of processing, 'slips of the tongue' errors, staged-child development, etc.).

Below, for now, are just the list of the four sentences (later discussed in Lecture-3). (Also see link below for additional discussion). For now, as a precursor of things to come, try to see if you can understand why such structures are important for child language studies.

Four Sentences:

- (i) Can eagles that fly swim?
- (ii) Him falled me down.
- (iii) The horse raced past the barn fell.
- (iv) I wonder what that is/*'s up there. (where clitic {'s} can adjoin to [that's] (note that all clitics theoretically become incorporated into the stem as [that's] and not *[[that]'s]. (e.g., phonologically, {that's} syllabically marks as a single word).

Even the simple example in (v) (first illustrated by Bickerton) shows how *linear adjacency* (of say noun with verb) *doesn't* apply in certain syntactic constructs:

- (v) The Boy Bill asked to speak to Mary thinks he is smart'. Who is thinking? Mary??

It turns out No, it's not Mary thinks— despite the two words being adjacent (and certainly these two words strung together such as 'Mary thinks' is very prevalent in our input). It turns out rather that 'recursive structure' dictates that the matrix noun to be processed with the verb 'thinks' is in fact way up in the sentence chain: 'The boy' is doing the 'thinking', not Mary or the next noun closest Bill.

The upshot here is that it seems the embedded structures-[[]] guides us to jump and advance in ways not seen if mere adjacency-[] ruled language processing.

Imitation>Analogy>Computational.

While there is robust evidence that children initially move from using both sides of their brain to using the left hemisphere, known as lateralization (e.g., Mills et al. 1997), the question has been— What is the nature of such incremental steps leading to full left-brain lateralization of language, and how might such incremental phases map onto our current understanding of syntactic theory?

Coupled with this, what current developmental linguists (see my paper in Journal of Child Language Acquisition and Development – see link below⁷) are looking for is a viable unifying processing model' which can account— incrementally in the child over time—for a triad processing of:

- (a) **imitative sound-to-meaning** conservative word mapping via non-productive rote-learning [+frequency-sensitive] (e.g., book > /bŪk/, tree > /tri/),
- (b) **analogies** which then give way to semi-productivity within word schemes [+frequency sensitive] (e.g., present to past tense zero-inflection {Ø} internal word schemes of verbs which end in /__t/: hurt>hurt-Ø therefore set>set-Ø, or the [#ing>#ang>#ung] analogy which produces correct sing>sang>sung, ring>rang>rung> but erroneous *bring>brang>brung), and finally
- (c) **a rule-based computational** system which defies sensitivity to frequency all together and is rather purely productive—for example, the productivity [-Freq(ueency)- sensitive] of a default phonological rule of such verbs ending with /__t/ to /Id/ (wanted, visited > to over-regularized *hurted). This latter (overregularized) processing speaks to the true abstract nature of rules and suggests movement operations such that a category , or morphological so by adding via default an {ed} to any category V to project past tense [[V] ed]. (By extension, Berko's classic _Wug's Test' (Berko 1958) is applicable: [[wug] -PI] becomes [[wug]s]).

Recall, that these three modes of processing represent the evolution of linguistic theory (from Skinner to Chomsky). It's easy enough to see imitation at work in language, children do repeat words they hear in the environment (but also not that direct X=X imitations schemes come under

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https://www.academia.edu/15155921/Small_Children_s_Sentences_are_Dead_on_Arrival_Remarks_on_a_Minimalist_Approach_to_Early_Child_Syntax_Journal_of_Child_Language_Acquisition_and_Development_vol_3_no_4_Dec_2015

pressure when children don't exactly repeat but rather alter their inputs, such as XY-input goes in by X-output, or XYZ-output comes out (omission and commission respectively).

Is quite interesting to see if experiments can be created which both support and/or violate such modes of processing. For instance, an overextension of ing >ang>ung might be primed with nonce words such as bling, gring, pling...ask a child to complete the sentence:

Experiment: Today I bling, yesterday I __, and I have __

(see if you get blang, blung).

Any putative underwriting of such incremental processing would look like the following, using variable formation.

Consider (i) imitation (of item) > to (ii) analogy (of sound pattern)> to true rule-based variable/categorical (iii) computation:

(i) **Imitation:** $[x + x \Rightarrow x]$ (lexicalization)

e.g., $[V \text{ break}] + [N \text{ fast}] = [N \text{ breakfast}]$,

$[N \text{ wine}] + [N \text{ bottle}] = [\text{AdjP wine bottle}]$

⇒ where X+X item-based memorization is employed.

(ii) **Analogy:** $[w [xy]] \Rightarrow [z [xy]]$ (analogy) $[_[ug]] > [_[ugs]]$ bug > bugs, $[wug] > [wugs]$

⇒ Where sound-pattern analogy is employed: e.g., 'ug' sounds like 'bug, wug=> marked by'

(iii) **Computational:** $\{\gamma \{\alpha, \beta\}\}$ or $x + y = z$ (computational)

e.g., two $[[\text{book}s]]$, two $[[\text{wug}s]]$ John $[[\text{drive}s]]$, has $[[\text{driv}]\text{en}]$

⇒ Where true viable categorical-rule manipulation not tethered to either imitation or analogy is employed. (Chomsky, via Berko's 'wugs test' emphasis this latter mode in (iii)).

'Fix' vs 'Speaks' & 'Cars' vs 'Carson'

For our last thought-experiment, consider the two pairs 'Cars' vs 'Carson', and 'Fix' vs. 'Speaks'. There are two linguistic structures we can look at here regarding whether or not language is like a clock, where you just count words like minutes, where sequence rules: where you count one item after another. If language and brain is devised as a 'complex clock', then we should expect similar results based on similar distributions. In other words, if all is flat, linear []-structures, then certain expectations follow...'count and process that which follows'. But as we have been discussing, there are certain 'gaps & jumps' that we find in language processing which suggests rule-based and categorical processing as noted by double embedded structures: [[]].

There are two modes of processing here we can look at:

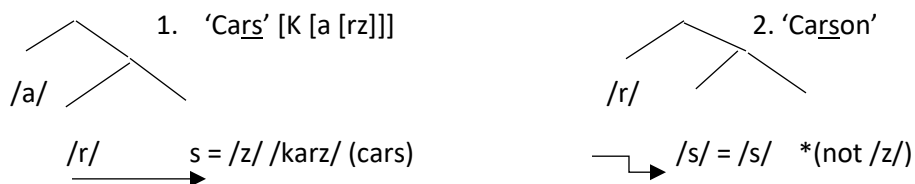
- (i) **assimilation**, a phonological processing—where one sound assimilates and wants to sound like its following sound. This would be a perfect example of language as clocks. The question to ask here is the following: Does the /r/ and /s/ in the word ‘Cars’, one sound which follows after the other, show similar results as compared to the /r/ and /s/ in the word ‘Carson’. As we will see, we can’t just count first comes /r/ and then comes /s/ as subsequent minutes in a clock...The fact that there are different outcomes of assimilation, as well will see, suggests that there must be different, albeit hidden, underlying structures.
- (ii) **stem+affix processing** as based on the sequence of two sounds which follow /k/ and /s/ as in the /...ks/ sequence as found in the two words ‘fix’ vs ‘speaks’.

Let’s consider the discussion below showing how hierarchy, which is a [[]]-recursive platform, shows up in the two examples below. We will be addressing the hierarchical distinctions as either *flat sisterhood relations* (where there is no hierarchy and where the sequence of language sounds/items are flat, as compared to *mother-daughter relations* (where there is hierarchy). In this brief recount below, compare the two types of structure (flat vs recursive) and see if you can account for the differences in outcomes between both the /r s/ sequence, and the /k s/ sequences as found in the pairs.

‘Cars’ vs ‘Carson’

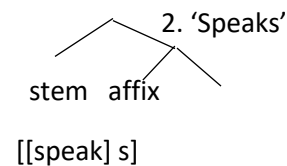
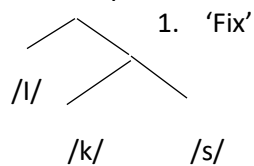
Whether it be... constraints placed on *phonological assimilation* which stipulate that in order for the *horizontal* spreading of voicing to occur between two adjacent consonants, they must first be of a ‘sisterhood’ relation; whereby, for example, the /r/ in ‘cars’ provokes assimilation of plural /s/ => /z/, in contrast to the /r/ in ‘Carson’ which does not. (The former a structural ‘sisterhood’ relation, the latter a ‘mother-daughter’ relation in terms of ‘family-tree’ hierarchy):

- (i) ‘Carson’ being broken-up by a *syllabic stress* /\$ /... /kar \$ sən/ (CVC \$ CVC).



Assimilation of ‘voicing’ applies between sisters. *No assimilation between mother-daughter. Or whether it be... the naïve view that the two apparently adjacent final-position sounds of /_ks/ as found in the two words ‘fix’ /flks/ versus ‘speaks’ /spi:ks/ surely must equally get processed similarly (they do NOT) calls on us to reconsider something much more insidious going on in the *underlying structure* of morphophonology:

(ii) 'Fix' /flks/ vs 'Speaks'



The differences in processing resulting in such distinctions can account for both developmental (as found in child language) as well as for Second Language (L2) errors of omission: viz., with a lexical stage-1 of child language often deleting the final 'mother-daughter' //s/ inflectional affix but never deleting the final 'sisterhood' //s/ stem-based element (with similar findings for L2 adults).

And then... to think that such *scaffolding* of 'mother-daughter' hierarchical structure which yields a *recursive syntax* comes to us for 'free'—part-and-parcel of the design of the human brain/mind—is something to wonder. This is what this book is about—'the wonder and unfolding of recursive syntax', and the manner in which it has forced the field of modern-day linguistics to reconsider old assumptions we once held dear—old assumptions which were hard to kill off, but which had to eventually die at the stroke of the *generative grammar enterprise*.

Just imagine, for any attempt to study early child language, prior to the Chomskyan revolution, it must have been very interesting, and of course problematic, to try to come up with any viable theory to account for the processing distinctions apparently found between the two adjacent sounds /ks/ in the two words—specifically, the question is: Why do children delete the final /s/ in 'speaks' (e.g., *Mommy speak, Daddy work*, etc. where young children delete the INFL affix (see lecture 5)) as compared to their never deleting the /s/ in other verbs such as 'fix'? Even though the spelling of 'fix' doesn't provide the /s/, there certainly is an /s/ in the speech. Both are verbs, both have identical distributions as found in the surface-level input ../ks/. Then, why the difference? Similarly, with the adjacent /rs/ sounds as found in 'cars' vs 'carson' (where we pronounce the /s/ as /z/ via assimilation in 'cars' (carz) but not so with the 'carson' (*carzon)?

As we have been exploring in this prologue, the notion of language as having a recursive, hierarchical status completely changes our understanding of language structure, and brought language into the modern era in terms of language as an algorithm, computational and recursive system.

Prologue Note: The evolution from items to categories

As a final prologue note, it's interesting to consider in evolutionary terms how this dual model dichotomy of 'clocks vs clouds' might have become the very seed to unraveling the true nature of our early-homo ancestors Cro-magnum in contrast to its closest competitor Neanderthal, some 40-60 thousand years ago. If Cro-magnum (*us*) came to the scene fully equipped with abstract and symbolic mental processes, it's easy to envision how the former sub-species (again, us!) might have eventually outwitted the latter in terms of communication, skills, tool-making capacity, organizational hunts and gatherings, and shelter. Imagine, if Neanderthal only came equipped with

clock-based ‘items’ (cued-representations) whereas Cro-magnum held in direct opposition to the latter the abstract ‘clouds’ of ‘furniture’ as a category:

[*items*: table- α , chair- β , nightstand- γ [*category*: furniture- δ]],

If Neanderthal were indeed stuck in a *flat* item-based world of processing whereby only cued-direct items could be linearly expressed, or possibly stacked on top of each other lexically, (of the ‘table-chair-and- nightstand’ variety), then there can be no doubt that Cro-magnon, drenched in their detached symbolic categories (furniture), would eventually come to outpace their rivals. Such a cognitive thought-process which allows its host to see the world via *categorization* would undoubtedly lead to the displacement of the earlier species. Modern Homo-sapiens have made the most out of this cognitive niche, up until today (and with artificial intelligence fast approaching—for better or for worse (though I have some thoughts on the topic)).

As a final note, perhaps the notion ‘ontogeny-recapitulates-phylogeny’ (ORP) is not too far off the mark, as is typically decried, at least not in these respects—where we do in fact find early states of child syntax initially taking-on a flat ‘beads-on-a-string’ methodology, a biased item-based processing completely reliant on the *frequency* of cued-based representation. However, what we do later find—via an emergent second, more formal stage—is the delivering of true *categorization* not sensitive to frequency. This ORP-theme seems to have played itself out accordingly, over the tens of thousands of years, within the emergence of our early-hominid human species only to be re-enacted once again in the narrow, ‘few-month-closing’ window of infancy by the maturing child—viz., where first-order recurrent structure quite rapidly gets surpassed by a maturation-based second-order recursive structure. This is the final act of our evolving human story: where homo-sapiens of today have found themselves at a place where they should rather be called *Homo-recursive*, or at the very least, *Homo-grammaticus*. It has a nice ring to it: ‘Homo-recursive’. Don’t you think? (My young abstract-art-loving daughter—who created the book-cover artwork and page-1 ‘duck-penguin insert’—wants to chime in with the suggestion *Homo-abstracticus*).